

Week 2 Day 4 Compose App Lab

Purpose

이 실습은 Day 3에서 길게 입력한 `docker run` 옵션을 `compose.yaml`로 옮긴다. 핵심은 파일 하나로 `web`, `db`, `network`, `volume`, `environment`, `healthcheck`, `cleanup`을 재현하는 것이다.

Setup

```
cd week2/day4/labs/compose-app
cp .env.example .env
```

`.env`의 `POSTGRES_PASSWORD`는 로컬 실습용 값으로 바꾼다. 실제 운영 `secret`이나 개인 `password`를 쓰지 않는다.

Run

```
docker compose config
docker compose up -d
docker compose ps
for i in 1 2 3 4 5; do
  curl -I http://localhost:18084 && break
  sleep 1
done
curl -s http://localhost:18084 | grep compose-site-v1
docker compose logs db
docker compose run --rm db-client
```

Expected Evidence

```
web-1    running
db-1     healthy
HTTP/1.1 200 OK
compose-site-v1
current_database | current_user
paperclip       | paperclip
```

Failure Drill

.env에서 POSTGRES_PASSWORD 줄을 제거한 뒤 config를 확인한다.

```
docker compose config
```

기대 결과는 set POSTGRES_PASSWORD in .env 메시지가 포함된 실패다.

Cleanup

```
docker compose down
docker compose down -v
```

docker compose down은 container와 network를 정리한다. docker compose down -v는 named volume까지 삭제하므로 DB data가 사라진다.

File Map

Path	Purpose
compose.yaml	web, db, db-client service 정의
.env.example	local .env 작성 기준
html/index.html	nginx가 제공하는 정적 페이지
db/init/001_schema.sql	PostgreSQL 최초 초기화 SQL
README.md	실행, 확인, 실패, 정리 기준

Services

Service	Image	Role	Exposed To Host
web	nginx:1.27-alpine	static HTML server	yes, \${WEB_PORT:-18084}:80
db	postgres:16-alpine	PostgreSQL database	no
db-client	postgres:16-alpine	query/check tool	no, profile tool

db-client는 기본 application service가 아니라 확인용 service다. 필요할 때 docker compose run --rm db-client로 실행한다.

Environment Variables

Variable	Required	Default	Notes
WEB_PORT	no	18084	host에서 web에 접근할 port
POSTGRES_DB	no	paperclip	practice database name
POSTGRES_USER	no	paperclip	practice user
POSTGRES_PASSWORD	yes	none	.env에서 로컬 값으로 설정

POSTGRES_PASSWORD는 required variable이다. 값이 없으면 docker compose config가 실패한다. 의도적으로 실행 전 실패하게 만들어 secret 누락을 빨리 발견한다.

Recommended Local Setup

```
cp .env.example .env
```

그 다음 .env를 열어 POSTGRES_PASSWORD 값을 로컬 실습용 값으로 바꾼다. 실제 개인 password, Docker Hub token, cloud access key를 쓰지 않는다.

Preflight

```
docker version
docker compose version
docker compose config
```

Expected:

```
services:
  db:
  web:
networks:
  app-net:
volumes:
  pgdata:
```

Detailed Runbook

1. Start

```
docker compose up -d
```

2. Check service status

```
docker compose ps
```

Expected signals:

```
web    Up
db     Up (healthy)
0.0.0.0:18084->80/tcp
```

3. Check web

```
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
```

Expected:

```
HTTP/1.1 200 OK
compose-site-v1
```

4. Check DB

```
docker compose logs db
docker compose run --rm db-client
```

Expected:

```
database system is ready to accept connections
current_database | current_user
paperclip        | paperclip
```

Network Model

The host reaches the web service through published port `localhost:18084`.

Containers inside the Compose network reach PostgreSQL through service name db.

From	To	Address
host browser/curl	web	localhost:18084
db-client container	db	db:5432
host terminal	db	not published by default

Do not document the DB container IP as the connection contract. Container IPs can change. Use service name db.

Volume Model

pgdata stores PostgreSQL data at /var/lib/postgresql/data.

Command	Result
<code>docker compose down</code>	removes containers and network
<code>docker compose down -v</code>	also removes named volumes

Use `down -v` only when resetting practice data. Do not use it as the default cleanup for data that must be preserved.

Failure Drill 1: Missing Env

Remove `POSTGRES_PASSWORD` from `.env`.

```
docker compose config
```

Expected failure:

```
set POSTGRES_PASSWORD in .env
```

Fix:

```
cp .env.example .env
```

Then set a local practice password again.

Failure Drill 2: Wrong Port

```
curl -I http://localhost:80
curl -I http://localhost:18084
```

```
docker compose ps
```

Interpretation:

- localhost:80 may fail because the published host port is 18084.
- localhost:18084 should return HTTP 200 when web is running.
- This is a host port issue, not an nginx container port issue.

Failure Drill 3: Stale Volume

If you edit db/init/001_schema.sql after the database has already initialized, the change may not appear. PostgreSQL init scripts run only during initial database creation.

Practice reset:

```
docker compose down -v
docker compose up -d
```

Warning: this deletes practice DB data.

Short RCA Template

```
## Compose App RCA
- Symptom:
- Failed command:
- Error excerpt:
- Category: config / port / env / network / volume / readiness
- Hypothesis:
- Fix:
- Recheck:
- Prevention:
```

Security Notes

- Do not commit .env.
- Do not put real passwords or tokens in .env.example.
- Do not paste Docker Hub tokens, MFA codes, cloud access keys, or personal credentials into terminal screenshots.
- Prefer placeholder values in documentation.
- Keep DB port unpublished unless the exercise specifically requires host DB access.

Handoff Section For Student README

```
## Docker Compose
### Setup
```

```
cp .env.example .env
```

```
### Run
```

```
docker compose config
docker compose up -d
```

```
### Check
```

```
docker compose ps
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
docker compose run --rm db-client
```

```
### Cleanup
```

```
docker compose down
```

```
Reset practice DB data:
```

```
docker compose down -v
```

Evidence Checklist

- [] `docker compose config` succeeded.
- [] `docker compose ps` shows web running.
- [] `docker compose ps` shows db running or healthy.
- [] `curl -I http://localhost:18084` returns HTTP 200.
- [] body contains `compose-site-v1`.
- [] `db-client` query returns `paperclip`.
- [] missing env failure is documented.
- [] cleanup command is documented.
- [] `down -v` data deletion warning is documented.

Scoring

Item	0	1	2
Config	not run	run only	output interpreted
Web	no check	status only	HTTP and body marker
DB	no check	healthy only	query evidence
Network	no explanation	network name only	service name db explained
Volume	no note	volume listed	data lifecycle explained
Security	secret exposed	warning only	.env.example and no secret exposure
Handoff	none	commands only	expected results and cleanup warning

Completion Statement

This Compose app is complete when another student can clone the files, create ``.env``, run ``docker compose up -d``, verify web and DB evidence, and clean up without accidentally deleting data they intended to keep.